

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Paper No. STA-201: Descriptive Statistics and Distribution Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Write a note on the use of a scatter diagram to obtain an idea of the extent and direction of the correlation coefficient. [#1]
- (b) If Y_1 , Y_2 , and Y_3 are uncorrelated variables each having the same variance, then obtain the correlation coefficient between $(Y_1 + Y_2)$ and $(Y_2 + Y_3)$. [#1]

Question 2.

- (a) Define intra-class correlation and also derive the expression for the intra-class correlation coefficient. [#1]
- (b) Describe the concepts of multiple and partial correlation coefficients. Show that the multiple correlation coefficient $R_{1.23}$ in the usual notations is given by $R_{1.23}^2 = 1 - \frac{\Delta}{\Delta_{11}}$. [#1]

Question 3.

- (a) Define the consistency of data for attributes and derive the necessary conditions of consistency up to three attributes. [#1]
- (b) Find the coefficient of correlation between the ages of the following mothers (X) and daughters (Y):

Mother's Age (X in years)	24	26	28	30	32	34
Daughter's Age (Y in years)	4	5	6	8	9	11

[#1]**Question 4.**

- (a) Prove that $|E(X)| \leq E(|X|)$, provided all expectations exist. [#1]
- (b) Calculate the mathematical expectation of the sum of points when rolling a fair die. [#1]

Question 5.

- (a) Show that $\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]$, where all notations have their usual meaning. [#1]
- (b) Define the moment generating function and state its important properties. [#1]
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Question 6.

- (a) Define the geometric distribution and prove that it satisfies the lack of memory property. If $\bar{X}$ is the sample mean of n i.i.d. random variables, show that $M_{\bar{X}}(t) = [M_X(t/n)]^n$. [#1]
- (b) If $X_i \sim \text{Gamma}(a, b_i)$ for $i = 1, 2, \dots, n$ are independent random variables, then prove that $\sum_{i=1}^n X_i \sim \text{Gamma}(a, \sum_{i=1}^n b_i)$. [#1]
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Question 7.

- (a) If X is a random variable and $E[X^2] < \infty$, then prove that $P(|X| \geq a) \leq \frac{E[X^2]}{a^2}$ for all $a > 0$. [#1]
- (b) Let the joint probability density function of random variables X and Y be:

$$f(x, y) = e^{-(x+y)}, \quad x > 0, y > 0$$

Find the following:

- (i) $P(X > 1)$
- (ii) $P(1 < X + Y < 2)$
- (iii) $P(X < Y \mid X < 2Y)$

[#1]

Question 8. Describe any **three** of the following in brief:

[#1]

- (a) Rank Correlation
- (b) Normal Distribution
- (c) Intra-Class Correlation
- (d) Uniform Distribution
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